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Latent obstacles in older adults' digital health participation: a community-based hybrid cluster analysis with natural language processing

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Abstract

Background The intensification of global aging has made older adults' health issues increasingly prominent. Although digital health tools play a crucial role in aging societies, older adults still face multi-dimensional potential obstacles that restrict their effective participation. Natural Language Processing (NLP) can efficiently analyze the data of older adults' feedback in digital health to uncover hidden patterns. Therefore, this study constructs a community-based hybrid NLP analysis method to reveal the deep obstacle mechanisms affecting the digital health participation of the elderly.

Methods In this study, interview data from 35 older adults were collected through semi-structured focus group interviews in five community health activities. After pre-processing the data, a document-term matrix was constructed using Term Frequency-Inverse Document Frequency (TF-IDF) vectorization. By integrating the three methods of Latent Dirichlet Allocation (LDA) topic modeling, BERTopic topic modeling, and SBERT with K-means clustering analysis, the semantic features and topic distribution within the text data are systematically analyzed. Among them, BERTopic (K = 4) has become the most effective clustering model in this study with high word-level consistency (0.5328), reasonable topic diversity (0.5583), and good document-level consistency (0.8467).

Results The research findings indicate that the potential obstacles affecting older adults' digital health participation are primarily the functional and design manifestations of deeper, socio-psychological, and social environmental latent barriers. These are clustered into four aspects: psychological barriers to digital technology use, clarity and usability of health information, a mismatch between technology provision and user needs, and user interface (UI) design and accessibility challenges. Older adults' relatively low digital literacy leads to fear and anxiety toward technology. In obtaining effective information, they face the issue of information overload and have doubts about the authenticity and reliability of the information. The lack of positive technology learning experiences and external support makes older adults prone to a sense of low self-efficacy. The uneven distribution of community resources and the limited technical support capabilities of family members further exacerbate the obstacles to older adults' digital health participation.

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Conclusion This study proposes a new method for in-depth exploration of the potential obstacles to older adults' digital health participation from a semantic level by combining multiple clustering analysis methods. It provides valuable references and technical guidance for subsequent related research.

Keywords Digital health, Older adults, Community, NLP, Cluster analysis

Introduction

The trend of global aging is intensifying, and the health issues of older adults have attracted increasing attention. Data shows that by 2050, the population aged 65 and above is expected to account for 16% of the global total population [1]. Against this backdrop, digital technology has the potential to promote health and prevent diseases [2]. With its convenience and high efficiency, digital health technology has become an important means for older adults to manage their health. Research indicates that digital health tools have significant advantages in chronic disease management [3], public health [4], telemedicine [5], and access to health information [6]. However, explicit issues such as the digital literacy gap [7], technology availability [8], and the digital divide [9] pose numerous challenges and obstacles to the elderly's participation in digital health. Although some elderly individuals are willing to try using digital technology, compared to younger people, they require more time to learn digital technology [10]. Due to a lack of digital literacy and the inability to access necessary equipment and internet connections, the elderly are often excluded from participating in digital health services [11].

Existing studies have primarily focused on the obstacles faced by older adults in using digital health technologies, such as difficulties in operating devices [12] and technological complexity [13]. These studies have devoted significant effort to designing simpler interfaces and conducting technology training tailored to the elderly to overcome these explicit obstacles. Despite these research efforts and practices targeting explicit obstacles, the acceptance and usage rates of digital health tools among older adults remain low [14]. This suggests that addressing only explicit obstacles is insufficient, and latent barriers, such as socio-psychological factors [15] and social environmental factors [16], may restrict the elderly's digital health participation and the use of digital technology to a greater extent. Previous research has shown that information avoidance [17], technophobia [18], and intergenerational communication barriers [19] are the main latent barriers to older adults' digital health participation. These obstacles not only affect older adults' acceptance of technology but may also lead to inefficiencies in their health management [14]. In addition, negative emotions towards digital health tools, such as confusion and anxiety, significantly reduce older adults' willingness to use them [20]. Therefore, identifying and addressing these latent barriers is crucial for improving the digital health participation of the elderly.

As an important part of older adults' lives, the environment and resources of the community have a significant impact on their digital health participation [21]. The community is not only the core space of older adults' daily life but also an important venue for them to obtain health information, technical support, and social interaction. Research has shown that community support [22] and community education [23] can significantly enhance their digital health literacy. For example, digital health training courses organized by the community can effectively reduce the technological barriers and help them better understand and use digital health tools [24]. At the same time, community support networks can alleviate loneliness and technophobia when they use digital health tools [25].

In addition, community resources, such as technical support centers and health consultation services, provide older adults with the necessary equipment and information support to help them overcome technological accessibility obstacles [26]. The differences and complexities in the community environment make the latent barriers that older adults face when participating in digital health diverse and complex. For instance, differences in social and cultural backgrounds also play a role. In areas that emphasize family support, they may be more likely to obtain technical help from family members, thereby reducing the impact of technophobia. In contrast, in social and cultural contexts that emphasize individual independence, the elderly may be more inclined to solve problems independently and are more likely to face difficulties due to technological complexity [27]. Given the complexity of the latent barriers affecting older adults' use of digital health technology, researching the impact of community environments and resources on the latent barriers to digital health participation is not only helpful for understanding the complexity of older adults' digital health participation but also provides a basis for designing targeted community intervention measures.

Although progress has been made in the research field of older adults' use of digital health technology and digital health participation, there are still the following limitations. Firstly, most studies focus on explicit obstacles, such as difficulties in device operation and technological complexity, with insufficient attention paid to latent barriers, such as psychological stress and socio-cultural factors [28]. Existing studies often use questionnaire surveys or experimental designs, which are difficult to comprehensively capture the latent barriers in older adults' digital health participation. Questionnaire surveys often rely on standardized questions and lack in-depth exploration of the latent barriers in older adults' actual living environment [29]. Therefore, this study starts from the community perspective, systematically analyzing the impact of community environments and resources on older adults' digital health participation, thus expanding the research scope. This study combines semi-structured focus group interviews and NLP clustering analysis techniques to deeply analyze the latent barriers to older adults' digital health participation from the community perspective, making up for the deficiencies of existing research and providing a scientific basis for designing targeted intervention measures. The technical and design challenges identified in this study are not merely explicit obstacles, but rather the concrete manifestations of these deeper socio-psychological and environmental limitations. Based on the above research objectives, this study proposes the following research questions:

1. What are the latent barriers and their semantic characteristics of the digital health participation of the older adults?
2. What are the semantic relationships and topic distributions between different obstacles?
3. How do community environments and resources affect the latent barriers in the older adults' digital health participation?

This study makes the following contributions. Firstly, it identifies and quantifies the potential obstacles in older adults' digital health participation, providing a new perspective and methodological framework for digital health research. While some of the identified barriers are known, the primary contribution lies in applying a community-based hybrid NLP methodology to systematically and quantitatively reveal the deep semantic structure and hierarchical relationships among these latent barriers, moving beyond anecdotal evidence or simple qualitative thematic descriptions. It provides a scientific basis for designing targeted interventions, promoting age-friendly design of digital health tools, and advancing health equity among older adults. Secondly, this study proposes a new method for analyzing the potential obstacles to older adults' digital health participation from a semantic level, providing a technical reference for subsequent research. The justification for using NLP clustering over traditional qualitative analysis is that this data-driven, hybrid approach offers greater objectivity and scalability for large textual datasets, enabling the extraction and quantification of subtle, deeply embedded socio-psychological concepts that manifest as technological barriers, which is more efficient than manual coding in this context [30]. Specifically, this study discovered four core obstacles that affect older adults' digital health participation: psychological barriers to digital technology use, clarity and usability of health information, a mismatch between technology provision and user needs, and UI design and accessibility challenges. These findings provide new insights into solving the deep-seated problems faced by older adults in the digital health domain.

Related work

Research on older adults' engagement with digital health has historically focused on explicit barriers, such as a lack of digital literacy and access to technology [31]. More recent research has shifted focus to latent barriers, which are often psychological or socio-environmental in nature and prove more difficult to overcome. Psychological factors include technophobia, anxiety about making mistakes, and low self-efficacy [25]. Socio-environmental obstacles encompass a lack of community-based technical support, inadequate social networks for learning, and communication gaps with younger family members [32]. Furthermore, design-related issues, such as poor UI design and the complexity of health information presentation, are increasingly recognized as significant barriers that stem from a fundamental lack of age-friendly design principles [33].

In recent years, significant progress has been made in applying NLP in the health domain, particularly in analyzing latent barriers and emotional tendencies [34, 35]. NLP enables extracting semantic features from large-scale text data, revealing hidden semantic patterns and topic distributions [36]. Among the topic modeling methods, LDA and BERTopic have been widely used for the semantic analysis of health text data. LDA is a classic generative probabilistic model that assumes documents are generated from topic distributions, allowing it to identify latent topics within the text [37]. BERTopic, based on the Transformer model and leveraging the semantic representation capabilities of BERT embedding, can capture the deep semantic features of text and automatically identify semantic groups [38].

Clustering analysis is an important method in text semantic analysis, used to identify potential groups or patterns within data [39]. K-Means is a classic clustering algorithm that achieves data grouping by minimizing intra-cluster distances. However, its performance is limited when dealing with high-dimensional nonlinear data [40]. BERTopic offers a solution: The Uniform Manifold Approximation and Projection (UMAP) and Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN)clustering algorithms integrated into BERTopic have demonstrated stronger capabilities in processing semantic text [41]. For example, Chai et al.

analyzed fear-related text from social media using BERTopic and HDBSCAN [42].

The combination of NLP and clustering analysis provides powerful tools for latent barriers research. The integration of LDA and K-Means can simultaneously achieve topic extraction and semantic grouping [43]. BERTopic, through semantic embedding and adaptive topic modeling, can better handle complex semantic data [44]. The HDBSCAN can capture local structures within high-dimensional semantic data, facilitating the intuitive display of semantic distributions. These techniques have shown great potential in latent barriers research and provide important technical support for this study.

Methods

Ethics approval

This study was approved by the Human Ethics Committee of Macao Polytechnic University (Approval No: HEA005-FCA-2024). All participants were required to sign an informed consent form before the interviews, ensuring their understanding of the study's objectives, procedures, and potential implications. All interview data were anonymized, and participants' personal information and identities were strictly confidential in the research reports.

Participant recruitment

Older adults were recruited through community-based health workshops organized in collaboration with local community centers by organizing health-related lectures on various health topics. These lectures were held at the community center and attracted participants who were interested in self-managing health. This approach allowed participants to naturally express their experiences and needs regarding health management.

Study design and procedure

Data collection was conducted in Macao and Guangzhou, China, targeting older adults through semi-structured focus group interviews to explore their needs and experiences with health technologies. The semi-structured format provided flexibility, enabling participants to share personal experiences and perspectives under the guidance of predefined questions. This ensured a comprehensive understanding of their genuine needs and expectations. The focus group format was specifically chosen over one-on-one interviews to encourage peer-to-peer discussion and debate among older adults, which often stimulates deeper reflection, validates shared experiences, and helps surface common community-level barriers that might not emerge in a private setting [45].

The interview outline was designed based on a two-step process: a review of existing literature on digital literacy and health management barriers, and alignment with the three research questions. The outline was structured around three main thematic areas: 1) Current health management practices and use of digital tools (e.g., 'What tools do you currently use for health?'); 2) Challenges and negative experiences with digital health technology (e.g., 'What makes you afraid or reluctant to use these tools?'); and 3) Social and community support for digital health (e.g., 'Where do you get help when facing technical issues?'). This structured approach ensured coverage of latent socio-psychological, information-related, and community context factors.

This ensured a comprehensive understanding of their genuine needs and expectations. Interviews were conducted at locations convenient for participants, with all sessions audio-recorded upon obtaining consent. The interviews were carried out by the first, second, and third authors, and all participants received an informed consent form before participation, outlining the study's purpose, procedures, and ethical considerations. After consenting, participants signed the form on-site. The interviews, lasting 30 to 60 min, included open-ended questions focusing on participants' experiences, needs, and expectations regarding the use of technology for health promotion. Follow-up questions were asked to explore participants' experiences in greater depth and clarify relevant details. Field notes were taken during each interview to document additional observations and reflections.

All interviews were transcribed verbatim, and participants were assigned pseudonyms to ensure privacy. Regarding the audio recordings, a comprehensive voice anonymization procedure was implemented. The original interviews were recorded using high-quality digital recorders, with the files saved in a secure, password-protected digital storage system accessible only to authorized research team members. Before any analysis, the audio recordings underwent a voice anonymization process. The transcribed texts were then anonymized and processed digitally for subsequent NLP-based analysis. Interrater reliability revealed substantial to perfect agreement (Cohen's $\kappa = 0.82$) between two independent reviewers for data selection.

Interviews in both Macao and Guangzhou were conducted primarily in Cantonese, the native language of the participants, to ensure comfort and detailed expression. Transcription was performed verbatim into Chinese characters. Subsequently, a standardized text segmentation tool for Cantonese was applied, followed by machine translation into English for the NLP analysis. The use of Cantonese segmentation was crucial to accurately capture idiomatic expressions and specific vocabulary unique to the dialect that reflect socio-psychological concepts. Regarding sample size, 35 participants across five community groups were sufficient for this hybrid methodology. Data saturation, defined as the point where no new themes or semantic patterns emerged from subsequent interviews [46], was achieved after the fourth focus group.

Data preprocessing

Text preprocessing

Before topic modeling and analysis, the transcribed text data underwent systematic preprocessing to remove noise and enhance model performance. The steps included:

(1) Stop word Removal: Eliminating common words (e.g., "the", "and") with minimal semantic contribution.
(2) Tokenization: Decomposing text into individual words or phrases.
(3) Lemmatization: Reducing words to their base forms (e.g., normalizing "study" to "learn").
(4) Rare Word Removal: Filtering out words with extremely low frequency to reduce noise.

The preprocessed data were transformed into a format suitable for semantic analysis, forming the foundation for subsequent modeling and interpretation.

TF-IDF

To convert text data into a mathematical representation, TF-IDF vectorization was employed to construct a document-term matrix [47]. TF-IDF quantifies word importance by balancing term frequency within documents and inverse document frequency across the corpus. The formulas are as follows:

$$TF - IDF\,(t,d) - TF\,(t,d) \cdot IDF\,(t)$$

Where:

TF(t, d) represents the term frequency of word *t* in document *d*, that is, the frequency at which word t appears in document *d*.

IDF(t) represents the inverse document frequency of word *t*, which is used to measure the importance of word *t* across the entire corpus. The calculation formula is:

$$IDF\,(t) - log\frac{/D/}{1 + /d \in D : t \in d/}$$

Finally, the generated document-term matrix serves as the input data for subsequent topic modeling.

Topic modeling

LDA

LDA was applied for topic modeling. As a generative probabilistic model, LDA assumes documents are generated from topic distributions, which in turn are derived from word distributions. The model estimates document-topic and topic-word distributions by maximizing the likelihood function:

$$P\,(w/d) - \sum_{Z=1}^{Z} P\,(z/d) \cdot P\,(w/z)$$

Where:

P(w|d): Probability of word w in document *d*.

P(z|d): Probability of document d belonging to topic *z*.

P(w|z): Probability of word w in topic *z*.

LDA-generated topics provided an intuitive representation of text data's thematic structure, supporting further analysis.

BERTopic

BERTopic, a Transformer-based method, leverages Sentence-BERT embedding [48] for semantic representation and combines clustering algorithms to extract semantic topics. Its core formula integrates semantic vectors and clustering:

$$\mathrm{T{=}Cluster(E)}\ ,\mathrm{E{=}}\left\{\mathrm{e_t}\right\}\begin{matrix}\mathrm{T}\\ \mathrm{t{=}1}\end{matrix}$$

Where:

T: Extracted topic set.

E: Semantic vectors generated by BERT embedding.

Cluster: Clustering operation to identify topics.

Comparison and selection of clustering methods

From 432 raw texts, 319 were retained after screening and preprocessing. TF-IDF-LDA, BERTopic, and K-Means were compared based on clustering performance and topic coherence. Evaluation metrics included:

(1) Interpretability: Clarity of thematic representation.
(2) Visualization Quality: Intuitiveness of reduced-dimension results.
(3) Feature Extraction: Ability to capture key terms and semantic groups.

We have summarized three different types of topic modeling methods, and the specific process is shown in Fig. 1. We will compare these three different topic modeling methods and select the one with the best performance for text clustering analysis.

Evaluation indicators

We use topic coherence and topic diversity to measure the clustering results of different natural language processing models. Topic coherence measures the semantic

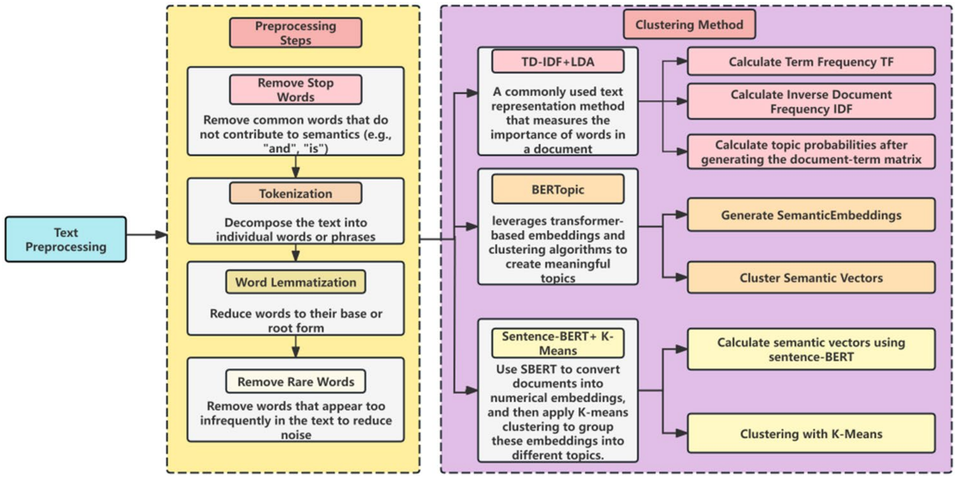

Fig. 1 Data processing and analysis process

consistency between keywords within a topic, assessing whether the keywords within a topic truly belong to the same semantic domain. Topic coherence is divided into document-level coherence and keyword-level coherence. Keyword-level coherence measures the semantic relevance or co-occurrence of topic keywords within the corpus. Topic diversity measures the degree of differentiation between different topics, specifically, whether the topics cover a sufficient number of different semantic domains.

Due to the different algorithmic mechanisms of the models, there are subtle differences in the calculations used by the three models. Table 1 lists the implementation strategies for these metrics.

In this study, we used radar charts to visualize the performance of topic modeling methods across three metrics: document-level consistency, keyword-level consistency, and topic diversity. To quantify the comprehensive performance of each model across these multi-dimensional metrics, we calculated the area of the closed polygon enclosed by each broken line in the radar chart as an indicator of the model's overall performance. A larger area indicates a better and more balanced overall performance across all metrics.

$$A=\frac{1}{2}\sum_{i=0}^{n-1} r_i\, r_{i+1} \sin\left(\theta_{i+1}-\theta_i\right), \quad \text{where } r_n=r_0,\ \theta_n=\theta_0$$

A: Area of the polygon, i.e., the closed region formed by the radar chart.

N: Number of metrics (e.g., document-level coherence, keyword-level coherence, topic diversity).

Value of the *i-t*h metric.

θ_i Angle corresponding to the *i-t*h metric (evenly distributed from 0 to 2π).

$r_n=r_0,\ \theta_n=\theta_0$ Close the polygon by connecting the last vertex to the first.

Results

Participant characteristics

This study included 35 older adult participants from Macao and Guangzhou, China, who were recruited through community-based initiatives to participate in community activities focusing on digital health

Table 1 Measurement indicators and testing methods for topic modeling models

Model	Document-level consistency (embedding-based)	Keyword-level consistency (word-level)	Topic diversity
TF-IDF + LDA (K = 5)	Use the average cosine similarity (KL divergence) between document-topic distribution vectors to measure intra-cluster tightness	Use the top-n topic words from LDA and compute keyword-level coherence with Gensim's *CoherenceModel* (c_v)	Use distinct-word ratio: count the number of unique words among all top-n topic keywords/total words
BERTopic (K = 4)	Based on Sentence-BERT document embeddings, directly compute the average cosine similarity of document embeddings within the same topic	Use keywords extracted via c-TF-IDF in BERTopic, then compute keyword-level coherence with *CoherenceModel* (c_v)	Use distinct-word ratio: based on top-n keywords from c-TF-IDF
SBERT + K-means(K = 8)	Clustering is directly based on SBERT document embeddings; compute the average cosine similarity of document embeddings within clusters	Use KeyBERT to extract keywords, then compute coherence with *CoherenceModel* (c_v)	Use distinct-word ratio: calculated on the set of extracted topic keywords

management and digital health technology education. Each participant was assigned a code that combined their identifier with a participant ID (e.g., E-1). Their ages ranged from 60 to 89 years old, with an average age of 80.6 years and a standard deviation of 6.73. The largest proportion of participants fell into the 70–79 age group (12/35) and the 80–89 age group (13/35), followed by the 60–69 age group (10/35). In terms of gender distribution, there were 24 females (69%) and 11 males (31%). Regarding living arrangements, 77% (27/35) of the older adults lived with their families, while 23% (8/35) lived alone. As for marital status, 68% of the older adults were married, and among those living alone, the majority were widows (9%) and divorced individuals (23%). The most common health problems were hypertension (64%) and diabetes (36%).

Determining the optimal topics for the model

Determining the number of topics (K) in topic modeling is a crucial step, directly impacting model quality and interpretability. Given the differences in the internal operating mechanisms of different clustering algorithms, we employed different methods for selecting optimal topics for each algorithm.

For BERTopic, since it includes automatic clustering, the clustering algorithm automatically selects the optimal number of topics. Therefore, we used a random generation method. We ran the algorithm 10 times with different random seeds (40–49). All 10 of these tests showed that the optimal number of clusters was 4. Therefore, we selected 5 clusters as the optimal number for the BERTopic model.

For TF-IDF-LDA, we used the elbow method to determine the optimal number of topics. The core idea of the elbow method is to find a balance between maintaining model quality and selecting the most concise and efficient number of clusters. We set a range of topic numbers (2 to 10) and trained a separate LDA model for each K value. For each model, we calculated topic coherence. We look for the "elbow" point where the growth rate of a relatively high consistency score begins to decline significantly. This point represents the point where the marginal benefits of increasing the number of topics begin to diminish, and the corresponding K value is considered to be the optimal number of topics for the model. Figure 2 shows the selection results of the elbow method. According to the topic consistency score, the highest point is at K = 8 (score: 0.4668). According to the slope change of the 'elbow method', a better candidate number of topics may be K = 5 (score: 0.3758). The slope change rate at the number of topics 5 is: 0.0758. Considering that the smaller the number of topics, the stronger the

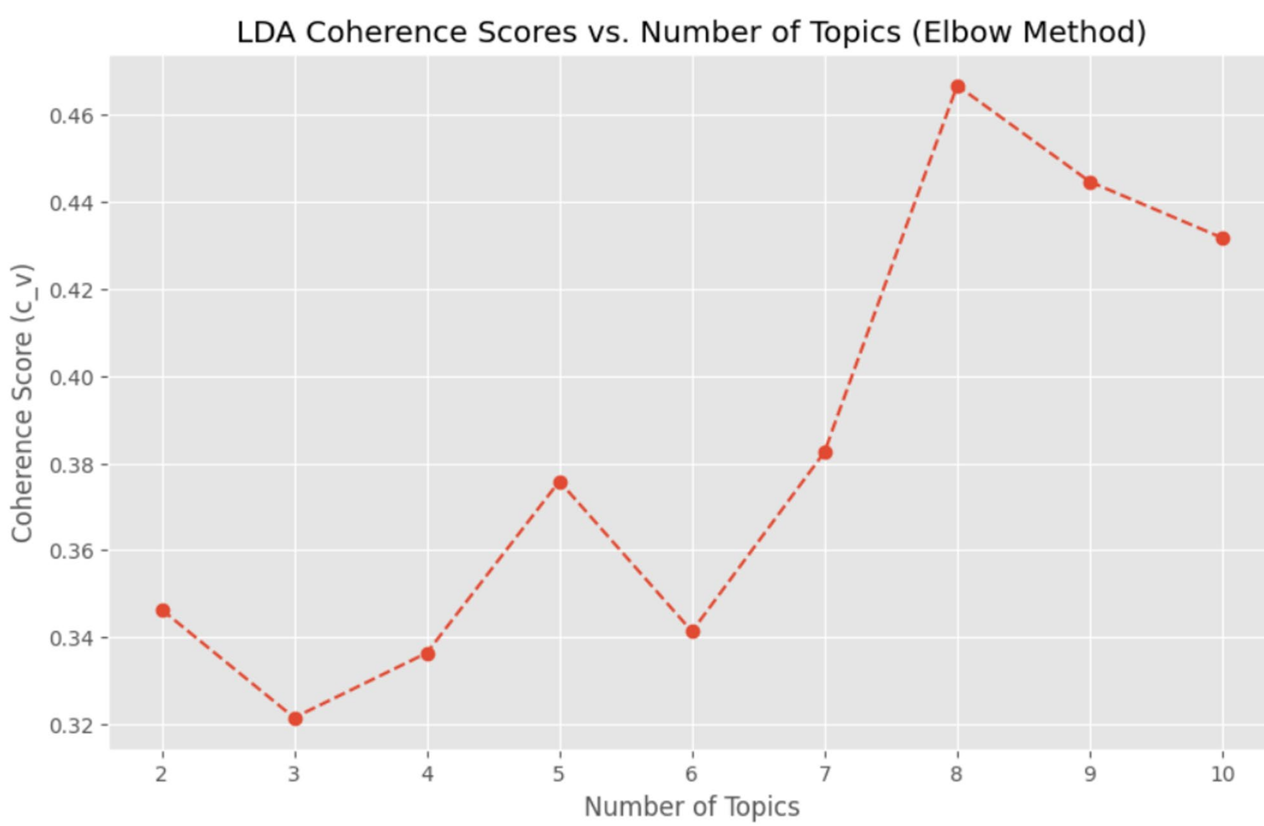

Fig. 2 Determine the optimal number of clusters K based on the coherence score

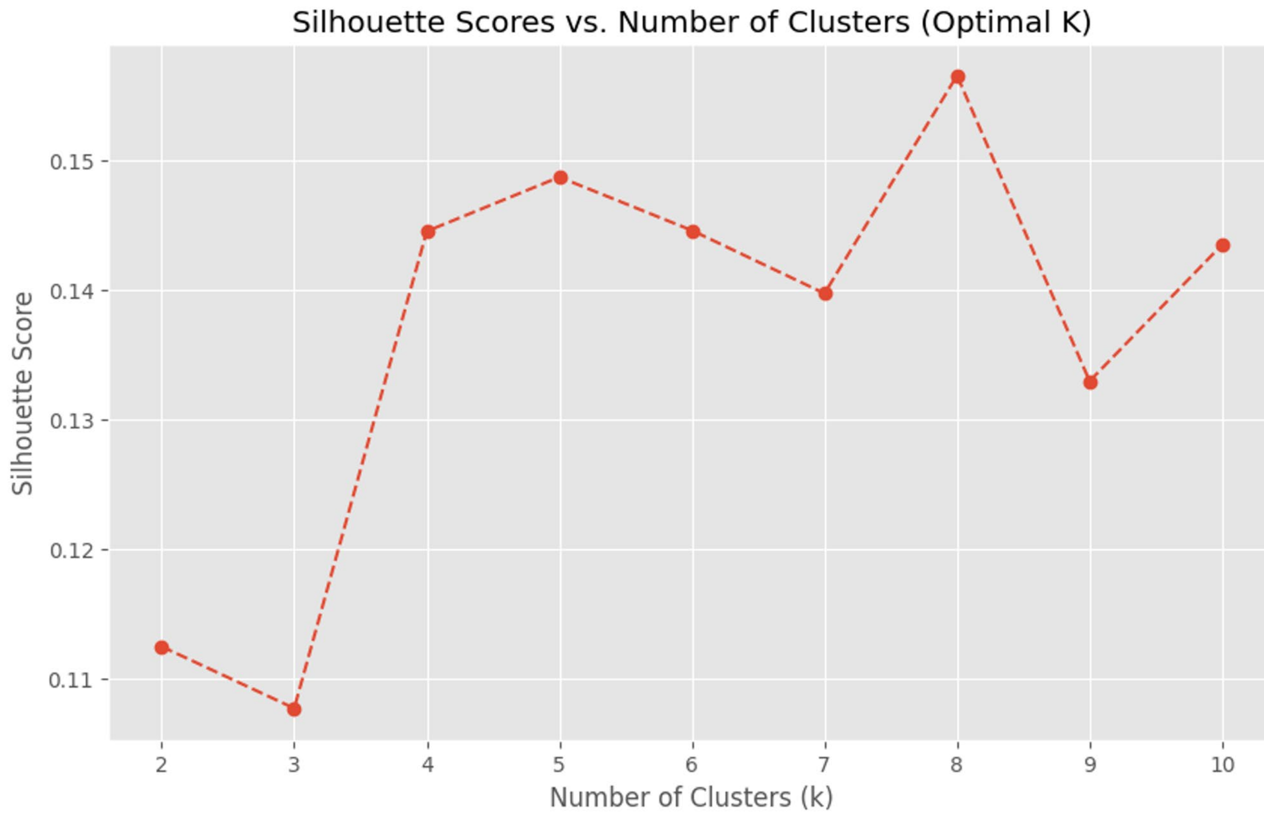

Fig. 3 Determine the optimal number of clusters K based on the silhouette score

Table 2 Test performance of topic modeling models

Model	Document-level consistency (embedding-based)	Keyword-level consistency (word-level)	Topic diversity
TF-IDF + LDA (K = 5)	0.9745	1.9745	0.675
BERTopic (K = 4)	0.8467	1.8467	0.5583
SBERT + K-means(K = 8)	0.6782	1.6782	0.7

interpretability, we choose the number of topics K = 5 as the optimal number of topics for the TF-IDF-LDA algorithm after comprehensive consideration.

For SBERT-k-means, we use the silhouette coefficient to determine the optimal number of topics. The silhouette coefficient (Silhouette Score) is a metric used to evaluate hard clustering algorithms. Unlike soft clustering algorithms such as LDA, the core feature of hard clustering is that each data point clearly and uniquely belongs to a specific cluster. The silhouette coefficient requires not only that the points within the cluster are sufficiently close, but also that the distance between different clusters is sufficient. The silhouette coefficient curve usually has a more obvious peak. The number of clusters corresponding to this peak is the point where both cohesion and separation are optimally balanced. Figure 3 shows the silhouette coefficient curve of the SBERT-K-Means algorithm. According to the Silhouette Score, the optimal number of clusters is K = 8 (Score: 0.1565), so we choose K = 8 as the optimal number of clusters for SBERT-k-means.

Model performance metrics

We measured the topic consistency and topic diversity of each model when K = 5. The results are shown in Table 2.

The indicators in the table show that different topic modeling methods perform differently across the three dimensions. TF-IDF + LDA achieves the highest document-level consistency (0.9745), indicating the tightest document clustering within a topic. However, it has the lowest keyword-level consistency (0.3364), indicating weak semantic coherence of topic terms. Its topic diversity is moderate (0.675), which is acceptable. BERTopic achieves a document-level consistency of 0.8467, slightly lower than LDA, but achieves the highest keyword-level consistency (0.5328), indicating the best semantic coherence of keywords extracted using c-TF-IDF. However, its topic diversity is the lowest (0.5583), indicating that the differences between topics are not as pronounced as with other methods. SBERT + K-means (K = 8) has the highest topic diversity (0.7), indicating that different topics are clearly distinguished, but the embedding-based consistency (0.6496) and keyword consistency (0.4102) are relatively low.

We calculated the area enclosed by the radar chart using the formula. As show in Fig. 4 BERTopic performed slightly better overall (0.5288), followed closely by TF-IDF + LDA (0.5251). The two methods had a good overall balance across the three metrics. SBERT + K-means had the smallest area (0.4436), indicating relatively weaker overall performance in document consistency, keyword consistency, and topic diversity. Therefore, we used BERTopic for subsequent analysis of this round of results.

BERTopic model analysis results

We used the BERTopic model to perform topic modeling on text data. First, the raw text was normalized and preprocessed, including punctuation removal, lowercase conversion, tokenization, stop word removal (based on the NLTK English stop word list), and removal of words with a length of less than 3. The processed text was used to train the BERTopic model. Document embedding was performed using the pertained Sentence-BERT model, all-MiniLM-L6-v2, encoding each document into a 384-dimensional vector. Vectorization was performed using CountVectorizer with parameters set to stop words = None and ngram_range= (1,1) to generate a document-to-word matrix. In subsequent steps, keywords for each topic were extracted using c-TF-IDF. To reduce the complexity of the high-dimensional embedding space, we used UMAP for dimensionality reduction with parameters: n_neighbors = 15, components = 5, metric='cosine', and random state = 40. The reduced vectors were used for subsequent clustering. The HDBSCAN algorithm is used in the clustering stage with the following parameters: min_cluster_size = 40, min_samples = None, cluster_selection_method='eom', to discover

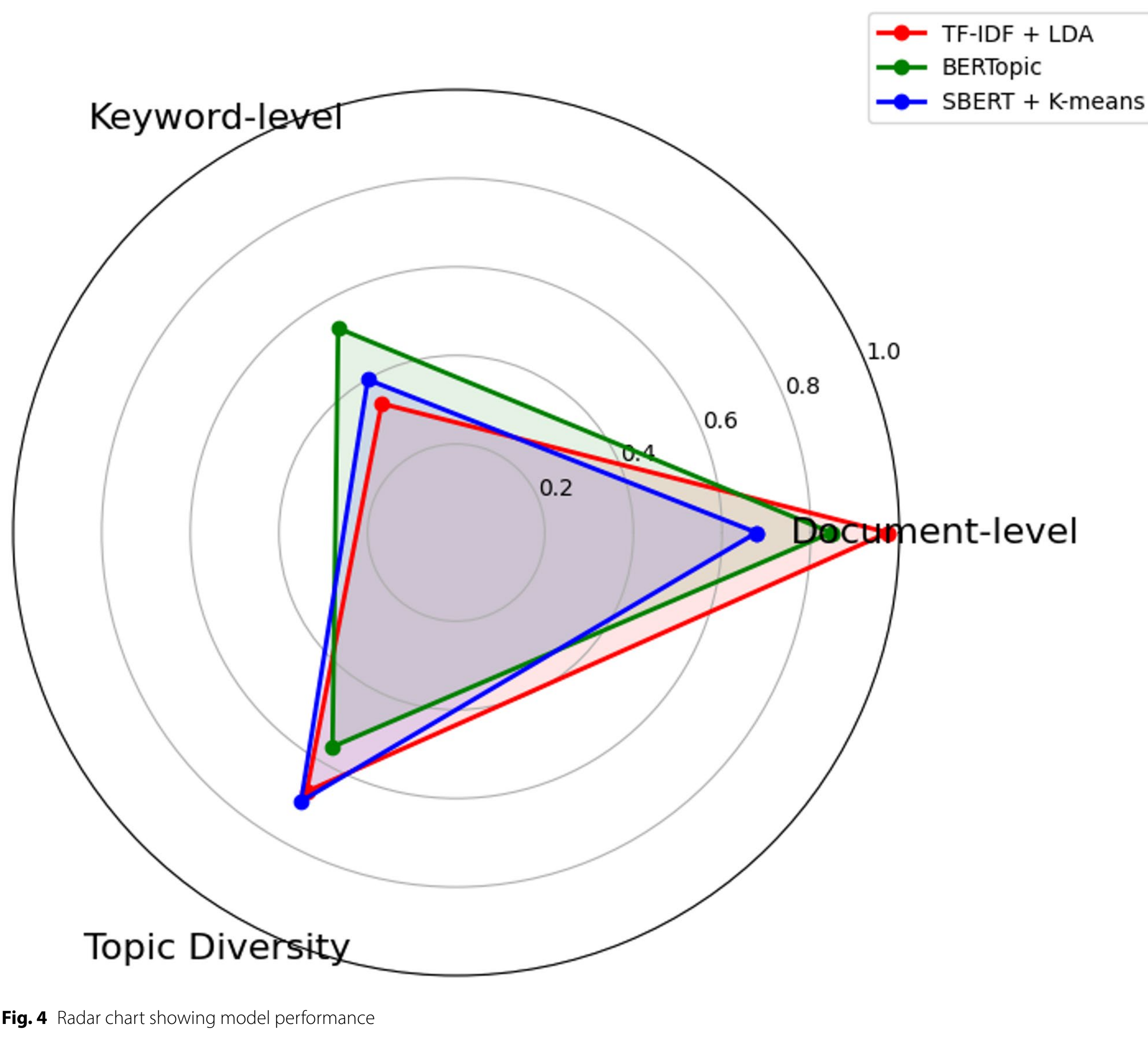

Fig. 4 Radar chart showing model performance

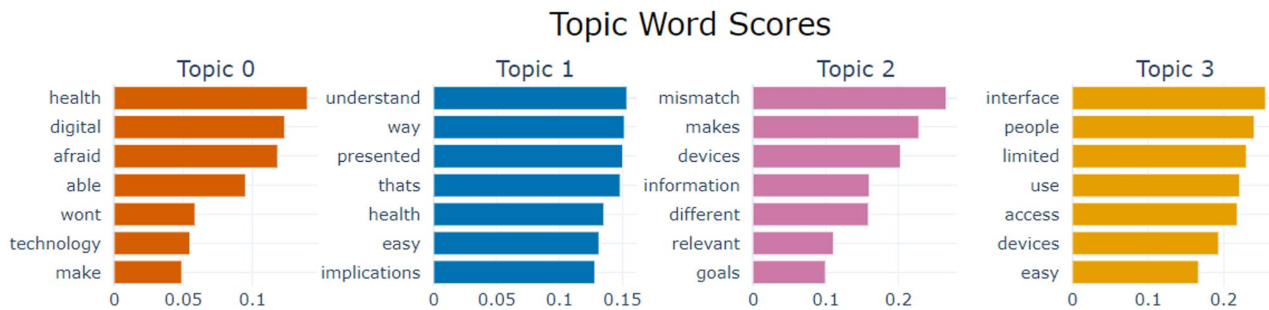

Fig. 5 Topic-Word Score Distribution Map

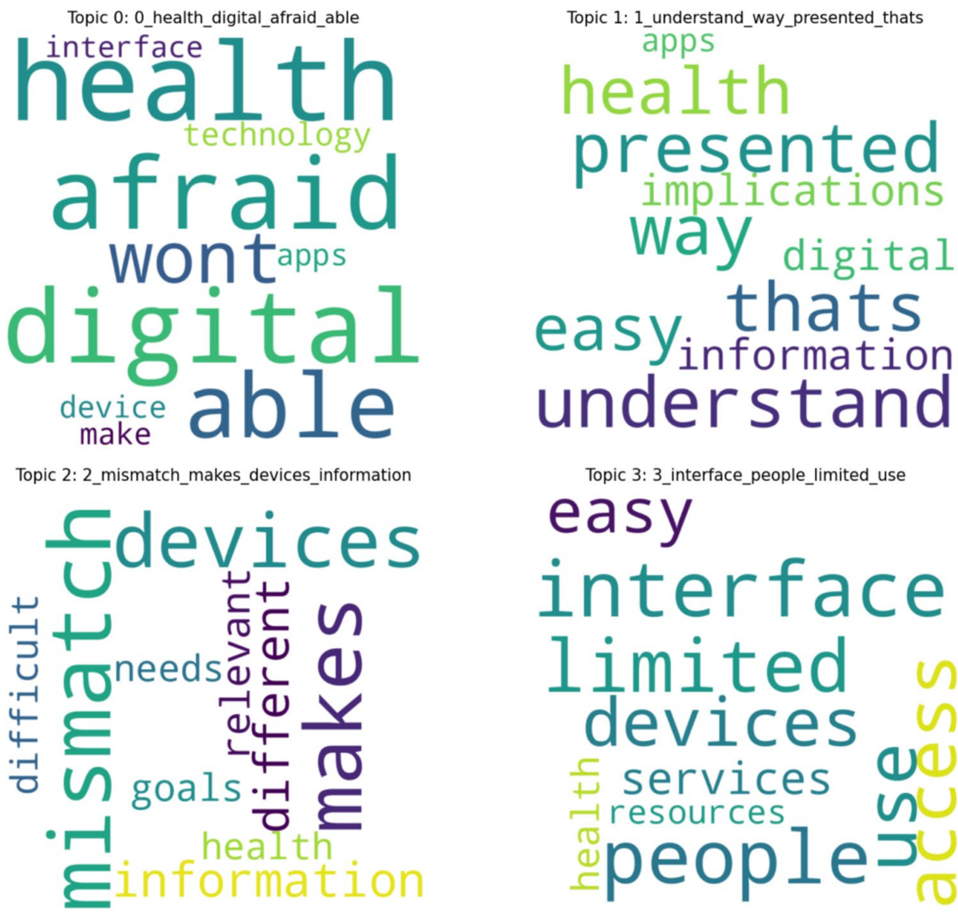

Fig. 6 Word cloud diagrams of various subjects

topic clusters in the document vector space after dimensionality reduction.

Based on a series of visualization charts generated by the BERTopic model, the thematic features and their internal associations within the text data are systematically presented from multiple dimensions. The topic-word score distribution map (Fig. 5) shows the core keywords and their importance for four topics (Topic 0, Topic 1, Topic 2, Topic 3). Figure 6 shows some word clouds related to the topic keywords. Table 3 shows a summary of the relevant topics and definitions.

For Topic 0, the core keywords are health (weight 0.1400), digital (weight 0.1234), and afraid (weight 0.1184). These terms indicate that this theme focuses on the psychological barriers to using digital health tools, specifically a fear of not being able to use the technology effectively. This clustering of terms reflects a user concern with whether they have the necessary skills or confidence to benefit from digital health services, highlighting a significant digital divide that limits access for certain groups.

Table 3 Summary of topic word clustering

Topic	Focus	Count	Description	Examples of prompt words for the classification topic
0	Psychological Barriers	127	Fear of technology, lack of confidence, and a fear of making mistakes.	"There is so much conflicting information about health on the digital device that I don't know which one to trust."
1	Information-related Obstacles	87	Difficulty processing vast amounts of information and discerning the authenticity of online health information.	"The information on the digital platform is not verified, and I'm afraid it's not reliable…."
2	Mismatched Needs	47	Digital health tools are often overly complex, failing to meet the elderly's need for simple, practical, and personalized health management.	"The digital health system's lack of integration with other health apps makes me afraid I'll have a fragmented health management experience…"
3	Age-friendly Design Challenges	42	Poor UI design that doesn't accommodate the decline in vision and cognitive abilities in older adults.	"The interface of the health device is so different from what I'm used to that it's hard for me to operate."

For Topic 1, the core keywords are understanding (weight 0.1536), way (weight 0.1516), and presented (weight 0.1503). These terms indicate that this theme focuses on the clarity and usability of health information within applications. This clustering of terms reflects a user concern with whether the data is easy to grasp and interpret, emphasizing the need for tools that don't just collect information but also transform it into meaningful and actionable insights.

For Topic 2, the core keywords are mismatch (weight 0.2658), makes (weight 0.2282), and devices (weight 0.2029). These terms indicate that this theme focuses on the incompatibility between a user's specific health goals and the technology's offerings. This clustering of terms reflects a user concern that the technology isn't personalized or relevant to their needs, and that data from different devices may be inconsistent, making it difficult to achieve a unified and relevant view of their health.

Topic 3, the core keywords are interface (weight 0.2555), people (weight 0.2405), and limited (weight 0.2301). These terms indicate that this theme focuses on challenges related to user interface design and overall accessibility. This clustering of terms reflects a user concern that the platform is not easy to use for all individuals, particularly those with limited access or ability, highlighting the critical importance of creating inclusive and intuitive designs.

The cosine similarity heat map in Fig. 7 further quantifies the degree of semantic association among five topics (Topic 0-Topic 4). The value on the main diagonal is 1, indicating that a topic is completely similar to itself. The depth of the color in non-diagonal cells reflects the level of similarity. For example, the similarity between Topic 0 and Topic 1 reaches 0.86, indicating a high degree of semantic correlation; the similarity between Topic 3 and Topic 4 is 0.56, indicating low association.

The hierarchical clustering map in Fig. 8 divides the topics into four categories through four horizontal lines. The small black dots at the ends of the lines represent the clustering merge points. Topics 0 and 1 merge at approximately 0.5–0.6 on the X-axis scale, indicating a relatively high degree of similarity between these two types of topics, while the distances between other categories are relatively large.

Discussion

Main findings

This study, by utilizing a hybrid NLP cluster analysis of interview data from older adults in a community setting, systematically identifies and characterizes four core latent obstacles to their digital health participation. These findings underscore that the challenges faced by older adults extend beyond simple technical literacy and are rooted in a complex interplay of psychological, cognitive, social, and design-related factors. Specifically, the four key obstacles identified are:

Psychological Barriers: This includes technophobia and a sense of low self-efficacy, driven by a lack of positive technology learning experiences and a fear of making mistakes. Information-related Obstacles: Older adults face challenges with information overload, the questionable authenticity of online health information, and the use of professional, non-user-friendly language in digital health tools. Mismatched Needs: There is a significant disconnect between the complex, feature-rich digital health tools designed by younger developers and the simple, practical, and personalized health management functions that older adults truly need. UI/UX Design Challenges: Older adults are hindered by poor UI design, including small fonts, low color contrast, illogical button layouts, and complex operational flows that do not accommodate their declining vision and cognitive abilities.

These findings collectively highlight the need for a more holistic and user-centered approach to developing digital health interventions for older adults, one that addresses not only technical accessibility but also

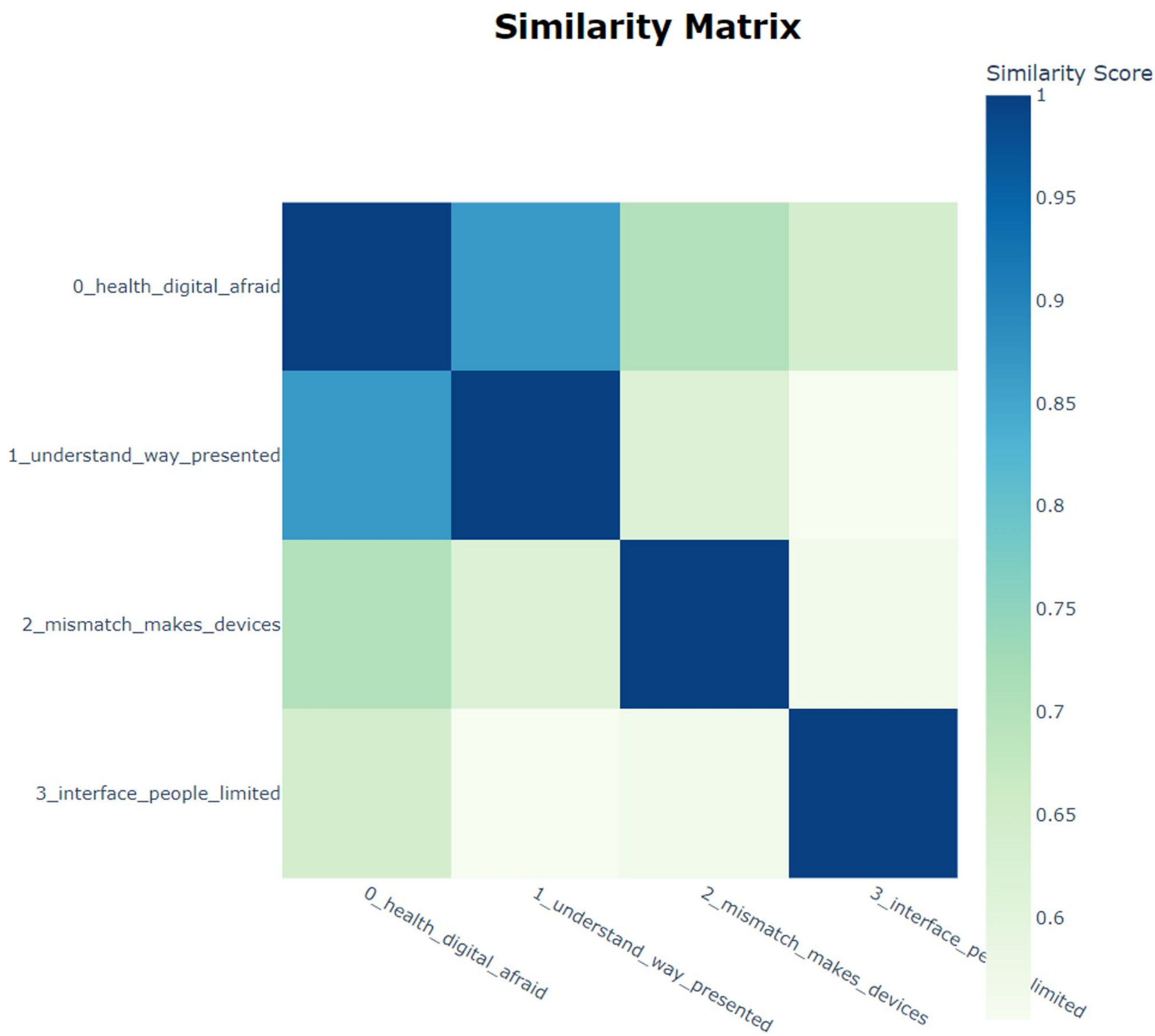

Fig. 7 Cosine Similarity Heat map

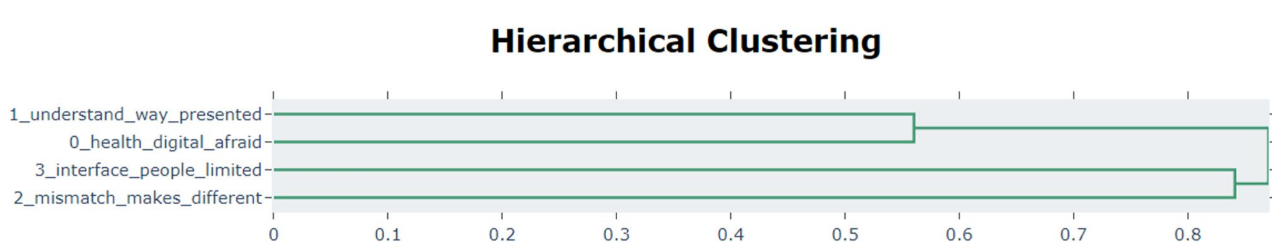

Fig. 8 Hierarchical clustering map

psychological readiness, information quality, and design-for-purpose principles.

Addressing latent barriers and the role of the community

Our findings suggest that while the four identified barriers (Psychological, Information, Mismatch, UI) are technical and functional in nature, they are fundamentally generated and sustained by deficiencies in the social and community environment. Linking Technical Solutions to Community Action: The optimization of UI, the design of digital health software functions, and the improvement of health information clarity are not solely technical fixes but must be driven, supported, and sustained within the community environment.

Community Support for Technical Solutions: For instance, improving UI/UX design requires community-led co-design workshops where older adults actively participate in feedback and testing, leveraging the community as a localized usability lab. Addressing Information-related Obstacles demands community health services to offer verified, culturally appropriate health information, using community centers as trusted physical hubs for digital content distribution and literacy training. The Mismatch of Needs can be mitigated by leveraging community networks to assess specific health management requirements and advocate for localized, simple digital tool provision. Finally, reducing Psychological Barriers is achieved by establishing supportive, intergenerational technical assistance programs within the community, normalizing technology use, and providing consistent, fear-reducing support. Thus, the community environment is not merely a setting, but the essential mechanism for implementing and supporting these seemingly technical solutions, directly influencing the reduction of latent barriers.

Psychological aspect: technophobia and operation anxiety

Technophobia and operation anxiety are prominent latent barriers in this study, which can be effectively explained through the lens of the Technology Acceptance Model (TAM) [49] and the self-efficacy theory [25]. TAM posits that perceived ease of use and perceived usefulness are crucial factors influencing an individual's intention to use a technology. In the context of digital health tools for older adults, the complex interfaces and multi-step operation processes lead to a low perceived ease of use. Many participants reported feeling nervous when operating digital health devices, fearing device malfunctions or data loss, as evidenced by statements like *"I get a headache just looking at the complex interface."* This fear stems from a lack of self-efficacy in their technological abilities [50], which is a core concept in self-efficacy theory. Self-efficacy refers to an individual's belief in their ability to perform a specific task. Older adults' lack of digital literacy, especially when faced with digital health tools, results in low self-efficacy, which in turn reduces their perceived ease of use and intention to use these tools [51].

The causes of psychological barriers can be attributed to two aspects. Firstly, older adults generally lack digital literacy, especially when faced with complex UI and multi-step operation processes; they are prone to frustration [52]. Secondly, technology design itself is not aging-friendly. For example, small interface fonts and overly complex operation paths may exacerbate older adults' fear of technology [53]. Research has shown that technophobia not only affects the acceptance of digital health tools among older adults but may also lead them to avoid using digital technologies [54].

To address these issues, we can draw on the principles of TAM and the self-efficacy theory. For example, providing operation guidance and step-by-step instructions, similar to the prompt function in short videos, can significantly enhance users' operation confidence. This is in line with the idea of increasing perceived ease of use in TAM. Emotional support tools, such as volunteer tutoring and sharing of success stories, can effectively reduce older adults' technophobia. These interventions can boost older adults' self-efficacy, as they provide positive role models and practical support, ultimately increasing their intention to use digital health tools. Studies on short-video learning have also indicated that providing operation guidance and step-by-step instructions (similar to the prompt function in short videos) can significantly enhance users' operation confidence [55]. This provides an important reference for designing digital health tools more suitable for older adults.

Health information acquisition: information overload and trust crisis

The information overload and trust crisis that older adults face when obtaining health information can be understood through the Health Belief Model (HBM) [56]. The HBM suggests that an individual's health-related behavior is influenced by their perception of the severity of a health problem, their susceptibility to it, the benefits of taking action, and the barriers to taking action [57]. In the case of health information acquisition, the large amount of information and doubts about its authenticity and reliability act as barriers.

Many participants stated that they often feel overwhelmed when browsing health information. This information overload is due to the lack of targeted distribution of health information, with older adults receiving irrelevant information that is difficult to filter [58]. The diversity of health information sources and differences in credibility exacerbate their doubts about the authenticity of information. Some participants mentioned seeing unverifiable health remedies on social media, leading to a loss of trust in health information [59].

To alleviate these issues, optimizing the information recommendation mechanism can be seen as increasing the perceived benefits of accessing health information. By pushing customized information according to older adults' health conditions and interests, the pertinence and understandability of information can be improved [60]. Strengthening the authority and credibility of health information, such as having medical institutions release health information, can enhance the perceived benefits and reduce the perceived barriers. Specific interventions could include developing a personalized health information push system. This system could use algorithms to analyze older adults' health data and send targeted health information via SMS or in-app notifications. Additionally, community-based health information workshops could be held bi-monthly.

Insufficient Self-efficacy: latent doubts about technological abilities

Insufficient self-efficacy is another type of latent barrier discovered in this study. Many participants expressed a lack of confidence in their technological abilities, believing that they could not master complex digital health tools. This lack of self-efficacy not only affects older adults' willingness to use digital health technologies but may also lead them to give up when facing technological challenges [61]. The causes of insufficient self-efficacy come from two aspects. Firstly, older adults lack positive learning experiences during the technology learning process, such as not being able to gain a sense of achievement from using technology. Secondly, there is a lack of external support for their technological abilities in the social environment, such as a lack of patient guidance from family members and insufficient support from community resources.

Research has shown that self-efficacy is closely related to older adults' technology use behaviors [62]. The lower the self-efficacy, the more likely older adults are to avoid technology. To address this, the design of digital health tools should incorporate instant feedback and positive reinforcement mechanisms. For example, when an older adult completes a step in using a digital health tool, they could receive a congratulatory message or a small reward. This helps them accumulate a sense of achievement in small steps and enhances their confidence in using technology. Measurable interventions could include creating a self-efficacy improvement program. This program could involve weekly online challenges related to digital health tool use.

Lack of support and resource acquisition barriers

The lack of social support and resource acquisition barriers are common latent barriers, which can be analyzed using the social support theory (SST) [63]. SST emphasizes the importance of social relationships and resources in an individual's well-being and behavior [64]. In the context of digital health, the uneven distribution of community resources and limited support from family members restrict older adults' access to digital health tools and social support.

The causes of a lack of social support mainly come from two aspects. Firstly, the uneven distribution of community resources means that many older adults lack the opportunity to access digital health tools [65]. Secondly, family members' technological support capabilities are limited, especially when they are busy with work or lack technological knowledge [66]. Differences in the quality and availability of community support in different regions and cultural backgrounds further limit older adults' digital health participation.

To alleviate these barriers, strengthening the investment in community resources is crucial. For example, providing technology support centers and health consulting services can improve older adults' accessibility to digital health tools. Measurable interventions could include establishing at least one community technology support center per 10,000 older adults, with each center providing at least 20 h of technical support per week. Family members' technological support capabilities can be improved through training courses.

In terms of policy, the government could promote the transparent management of health information, formulate information dissemination norms, and ensure that older adults obtain authoritative and reliable health information. Policy-makers could allocate a specific budget for community digital health support programs and evaluate the impact of these programs on older adults' digital health participation through annual surveys.

Age-friendly design and policy recommendations

At the design level, aging-friendly step-by-step guided operation processes should be provided to reduce older adults' fear of technological operations. This is in line with increasing the perceived ease of use in TAM and enhancing self-efficacy. The information recommendation mechanism should be optimized using personalized algorithms to push authoritative health information, improving the pertinence and understandability of information. Information summary functions can help older adults quickly obtain key information, alleviating information overload and trust crisis [67].

Regarding the problem of insufficient self-efficacy among older adults, the design should incorporate instant feedback and positive reinforcement mechanisms to help them accumulate a sense of achievement in small steps and enhance their confidence in using technology [68]. In terms of community support, tool design should embed health management courses and interactive functions to make up for the sense of isolation caused by the lack of social support among older adults.

At the policy level, it is recommended to strengthen the construction of community health support centers, provide technical training and equipment support, and guide family participation in older adults' technology learning through policies to alleviate the lack of social support. The government should promote the transparent management of health information, formulate information dissemination norms, and ensure that older adults obtain authoritative and reliable health information. In terms of community health center funding models, a specific percentage could be allocated for digital health support programs. Through the coordinated promotion of design improvements and policy support, the inclusiveness and universality of digital health technologies for older adults will be significantly enhanced, providing strong support for healthy aging.

The four barriers fit well with self-efficacy and technology-acceptance mechanisms. For example, technophobia leads to lower perceived control over digital health tools, which in turn reduces the intention to use them. Community support can enhance self-efficacy, as it provides the necessary resources and encouragement for older adults to use digital health tools. Each topic maps to specific constructs in these theories, providing a clear theoretical grounding for the identified barriers and proposed interventions.

Strengths and limitations

This study has achieved certain results in uncovering the latent barriers to older adults' digital health participation. However, several limitations need to be pointed out. Firstly, to justify our sampling strategy, we intentionally selected participants from both Macao and Guangzhou to capture diverse socioeconomic and cultural contexts. The sample remains urban-focused, spanning varying digital health infrastructures, income levels, and health system navigation patterns. We acknowledge limitations in rural representation and recommend future studies incorporate tier-2/3 cities and villages to test the robustness of these patterns, but argue the current sample provides meaningful insights into urban digital health adoption challenges in rapidly aging societies like China.

Future research can further verify and expand the findings of this study by expanding the sample size and covering more regions and cultural backgrounds. Secondly, the methodological limitation lies in the fact that although BERTopic performs well in semantic feature extraction and cluster analysis, its dependence on high-frequency keywords may lead to the neglect of some low-frequency but important latent barriers. Some barriers with less mention but significant socio-psychological impacts may not be fully captured. Therefore, future research can combine more complex NLP methods or multi-level semantic analysis techniques to further improve the ability of semantic feature extraction and the comprehensiveness of analysis. In addition, this study is a static analysis and does not consider the dynamic changes in older adults' digital health participation. Latent barriers may evolve, with changes in the technological environment and personal experiences. Future research can use a longitudinal study design to track the dynamic process of older adults' digital health participation and reveal the evolution rules of latent barriers in different periods and their driving factors.

Conclusion

In this study, we systematically revealed four core latent obstacles affecting older adults' digital health participation through a hybrid NLP cluster analysis of focus group interview data. These obstacles are: psychological barriers to digital technology use, clarity and usability of health information, a mismatch between technology provision and user needs, and UI design and accessibility challenges. These findings underscore that the obstacles are not merely technical issues but are the result of a complex interplay of psychological, cognitive, social, and design-related factors. To effectively address these issues, future research and practical efforts should focus on several key areas. Firstly, it is essential to develop more inclusive digital health products designed with older adults' actual needs and cognitive characteristics in mind, adopting large fonts, high contrast, clean and simple UIs, and simplified operational flows. Secondly, support at the community and family levels should be strengthened through initiatives like technology training and one-on-one guidance to help older adults overcome technophobia and low self-efficacy, thereby improving their digital health literacy. Lastly, future research can further explore the differences in obstacles faced by older adults from varying cultural backgrounds and health conditions and develop more universally applicable intervention models. Future studies could expand the sample to include a wider range of geographical areas to validate our findings and provide a more robust scientific basis for the development of digital health in the context of global aging.

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Authors' contributions

T.L. and Y.L. proposed the idea to conceive and design this study. Y.L. analyzed the data. T.L. completed the manuscript with the assistance of Y.L. P.P., D.W., and Y.W. supervised students and oversaw this work. All authors discussed the results and contributed to the final manuscript.

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Data availability

The data used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This human ethics application for this research has been approved by the Macao Polytechnic University (project ID: HEA005-FCA-2024), and conducted in accordance with the Declaration of Helsinki. All participants voluntarily participated and provided informed consent. Their responses were anonymous and treated as strictly confidential.

Consent for publication

Not applicable.

Competing interests
The authors declare no competing interests.

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